

Solving the Minimum-Impact Facility Location Problem

A computational study of an obnoxious facility location model

Daniel Garoz Vazquez

July 2026

AMOD – Università degli Studi di Roma Tor Vergata
Erasmus student, A.A. 2025–26

Obnoxious facility location

Facility location problems usually assume that **customers want to be close** to the facility (shops, warehouses, hospitals).

For some facility types the opposite holds:

- landfills, incinerators
- chemical / nuclear plants
- prisons, high-voltage lines
- wind turbines, adult venues

The population wants to be **as far as possible** from them.

Starting point of this project:

survey of Church & Drezner (2022) in *Computers & Operations Research*, 138:105468.

Specific problem chosen:

Minimum-Impact Location Problem
(Church & Cohon, 1976, Brookhaven Nat. Lab.)

— the historical first obnoxious ILP model, described in Sec. 2 (eqs. 1–5) of the survey.

The Minimum-Impact Location Problem

Data. n communities with populations a_i , m candidate sites, Euclidean distances d_{ij} , impact radius S , p facilities to open.

$N_i = \{j : d_{ij} \leq S\}$ = “dangerous” sites for community i .

Variables. $x_j \in \{0, 1\}$: site j left vacant. $y_i \in \{0, 1\}$: community i is impacted.

ILP formulation:

$$\begin{array}{ll}\min & \sum_i a_i y_i \quad \text{(minimise impacted population)} \\ \text{s.t.} & \sum_j x_j = m - p \quad \text{(exactly } p \text{ facilities opened)} \\ & x_j + y_i \geq 1 \quad \forall i, \forall j \in N_i \quad \text{(covering)} \\ & x_j, y_i \in \{0, 1\}\end{array}$$

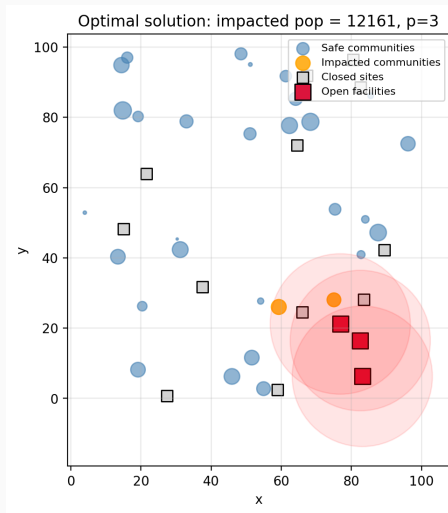
Number of vars: $m + n$. Number of covering constraints: $\sum_i |N_i|$ (grows with density).

Random instance model

Every random instance is fully determined by $(n, m, p, S, \text{seed})$:

- community and site coordinates: uniform in $[0, 100]^2$
- populations $a_i \sim \mathcal{U}[100, 10\,000]$
- distances d_{ij} Euclidean
- sets N_i derived from S

Fixed seeds $\Rightarrow$ 100% reproducible results.



Optimal solution on a small instance:

$n=30, m=15, p=3, S=20$,
impacted population = 12 161.

Solution methodology

Two-stage protocol per instance:

1. Solve the **LP relaxation**: z_{LP} is a valid lower bound.
2. Solve the **full MILP** with 120 s time limit: z_{MILP} is an upper bound (optimum if solver terminates in time).
3. Report the integrality gap $gap = 100 \cdot \frac{z_{MILP} - z_{LP}}{z_{MILP}} \%$.

Greedy approximation baseline:

open facilities one at a time, at each step choose the site with the **smallest marginal increase** in impacted population.

- Complexity $O(p \cdot m \cdot n)$
- Deliberately simple: no local search, no look-ahead

Tools. Python + PuLP + CBC (open-source), on Google Colab.

Experimental design

Two experiments, all with a 120 s solver time-limit:

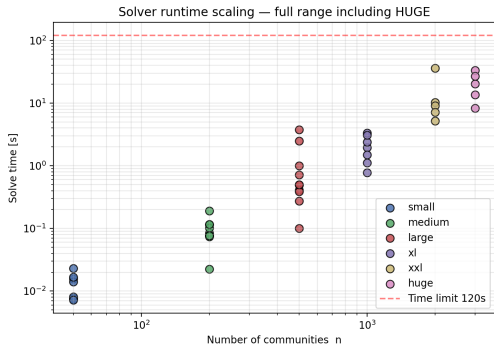
(A) Scalability study — 45 instances across 6 size classes:

Class	n	m	p	S	instances
small	50	25	4	15	8
medium	200	100	15	10	10
large	500	250	25	8	10
xl	1000	500	50	6	7
xxl	2000	1000	100	5	5
huge	3000	1500	150	4	5

(B) Sensitivity study — 15 instances at fixed size ($n=1000$, $m=500$, $p=50$), varying $S \in \{4, 6, 8, 10, 12\}$.

Total: 60 random instances.

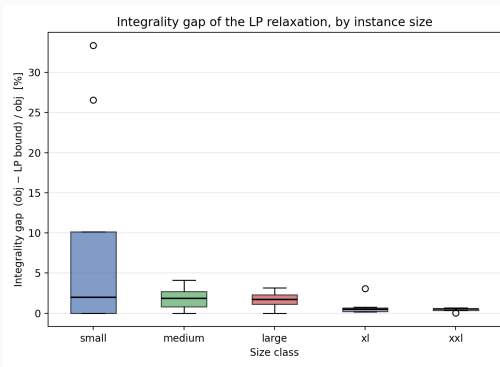
Scalability results



- 45/45 solved to proven optimum within 120 s.
- Average solve time: 12 ms (small) $\rightarrow$ 20.5 s (huge).
- Worst individual: 36.2 s (xxl).
- Nearly linear log–log scaling.

Takeaway. The MILP of Church & Cohon is remarkably solver-friendly under CBC across three orders of magnitude in n .

Integrality gap analysis

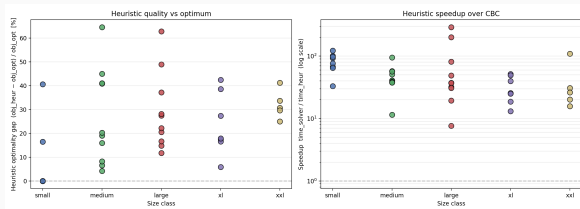


Counter-intuitive finding: the integrality gap of the LP relaxation *decreases* as instance size grows (8.6% $\rightarrow$ 0.4%).

Explanation. Larger n with proportionally scaled parameters $\Rightarrow$ denser covering matrix $\Rightarrow$ LP polytope closer to integer hull.

Branch-and-bound tree is very shallow (typically 1–2 nodes even in the largest classes).

Greedy heuristic vs CBC solver



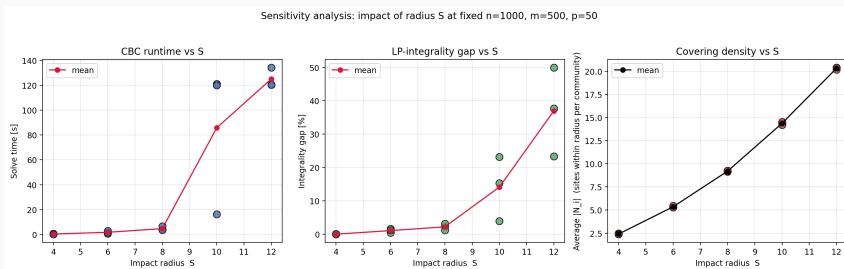
- Heuristic runs in tens of milliseconds even on the largest instances.
- Average **speedup**: $24\times$ – $290\times$.
- But average **quality gap**: 22–32%.

Verdict on the scalability instances:

CBC is so fast that the greedy is rarely worth the quality penalty.

Does the picture change at higher difficulty?

Sensitivity on the impact radius S



Fixing $n=1000$, $m=500$, $p=50$ and varying S :

- Difficulty cliff between $S = 8$ and $S = 12$
- Solve time: 4.6 s $\rightarrow$ 85.9 s $\rightarrow$ 125 s (first TLE!)
- Integrality gap: 2.2% $\rightarrow$ 37%

And the heuristic:

- At $S = 10$: gap drops to 7.6%
- At $S = 12$: gap drops to 3.9%

Difficulty is driven by covering density, not raw size.

The heuristic becomes competitive *in quality*, not just in speed, exactly where the solver breaks.

Conclusions

1. Church & Cohon's MILP is solver-friendly.

All 45 scalability instances up to $n=3000$ solved to proven optimum within 120 s.

2. Difficulty is controlled by covering density, not by n .

Time-outs appear only when the impact radius S pushes the average $|N_i|$ above ≈ 14 .

3. Solver and heuristic are complementary.

CBC is unbeatable in easy regimes; the greedy heuristic becomes competitive in both speed and quality precisely in the hard regime.

Reproducible Colab notebook available as supplementary material.

Thank you — questions?